EASY:

1. Write a program to store information of n students (Roll No, Name, Marks) in structure and display it.

Sample Input :
Enter No.Students: 1
Enter student 1 Roll No., name, Marks: 2001, AAA, 98

Sample Output:
Student 1 details:
Roll no.: 2001
Name: AAA
Marks: 98

Test Cases:

No.Student :4 (Any details of student)
No.Student: 5
No.Student: 1(2005, 4563, 62)
No.Student: A
No.Student: 1(xxx, John, 98.2)

2, Write a program to input and print student data using pointer object of structure.

Sample Input :
Enter No.Students: 1
Enter student 1 Roll No., name, Marks: 2001, AAA, 98

Sample Output:
Student 1 details:
Roll no.: 2001
Name: AAA
Marks: 98

Test Cases:

No.Student :4 (Any details of student)
No.Student: 5
No.Student: 1(2005, 4563, 62)
No.Student: A
No.Student: 1(xxx, John, 98.2)

3. Write a C Program to print the book details(Title , author, Subject,bookid - s structure member) using structures as Function Arguments
Sample input:
Enter the number of books: 1
Enter the book 1 title : C Programming
Enter the book 1 Author: Nuha Al
Enter the Book 1 subject : C Programming Tutorial
Enter the Book 1 book_id : 6495407

Sample output:
Book 1 title : C Programming
Book 1 author : Nuha Ali
Book 1 subject : C Programming Tutorial
Book 1 book_id : 6495407

Test Case:

No.books :4 (Any details of books)
No.books: 5
No.books: 1(2005,aaa ccc, 62)
No.Student: A
No.Student: 1(xxx,123,yyyy 98.2)

4. Write a C Program to Calculate Total and Percentage marks of a student using structure.
Input:
Enter RollNo,: 12
 Name : rama
 three sub marks 30 40 50

 Output:
rollno=12
Name=rama
m1=30 m2=40 m3=50 total=120 per=40.000000

Test Case

24, AA, 77,65,75
22, BB, 56,62,50
23,CC. 42,45,38
25, Annie 55,65,74
26, joy. 85,84,87

5. Write a program to store information of n students (Roll No, Name, Marks) in Union and display it.
Sample Input :
Enter No.Students: 1
Enter student 1 Roll No., name, Marks: 2001, AAA, 98

Sample Output:
Student 1 details:
Roll no.: 2001
Name: AAA
Marks: 98

Test Cases:

No.Student :4 (Any details of student)
No.Student: 5
No.Student: 1(2005, 4563, 62)
No.Student: A
No.Student: 1(xxx, John, 98.2)

6. Write a C program that uses functions and structures to perform the addition of complex numbers

Input:
Enter First complex number:
"enter real part of complex number: 2
enter Imaginary part of complex number:3

Enter Second complex number:
"enter real part of complex number: 4
enter Imaginary part of complex number:5

Output:
Addition of complex number :
Real part of addition : 6
Imaginary part of addition : 8

Test Cases:

2+5i, a+4i
3+6i. 6+4i
8+4i.12+7i
15-4i. 6-2i
3-8i.18-6i

7.Write a program to display information of n students (Roll No, Name, Marks) using Union
Sample Input :
Enter No.Students: 1
Enter student 1 Roll No., name, Marks: 2001, AAA, 98

Sample Output:
Student 1 details:
Roll no.: 2001
Name: AAA
Marks: 98

Test Cases:

No.Student :4 (Any details of student)
No.Student: 5
No.Student: 1(2005, 4563, 62)
No.Student: A
No.Student: 1(xxx, John, 98.2)

8. Write a C program that uses functions and structures to perform the Subtracting complex numbers

Enter First complex number:
"enter real part of complex number: 2
enter Imaginary part of complex number:3

Enter Second complex number:
"enter real part of complex number: 4
enter Imaginary part of complex number:5

Output:
Subtraction of complex number :
Real part of Subtraction : -2
Imaginary part of Subtraction: -2

Test Cases:

2+5i, a+4i
3+6i. 6+4i
8+4i.12+7i
15-4i. 6-2i
3-8i, 8-6i

9. Write a C program that uses functions and structures to perform the following operations:
iii) Multiplication of two complex numbers

Enter First complex number:

"enter real part of complex number: 2
enter Imaginary part of complex number:3

Enter Second complex number:
"enter real part of complex number: 4
enter Imaginary part of complex number:5

Test Cases:

2+5i, a+4i
3+6i. 6+4i
8+4i.12+7i
15-4i. 6-2i
3-8i.18-6i

10. Write a C program that uses functions and structures to perform the following operations: . Division of two complex numbers

Enter First complex number:
"enter real part of complex number: 2
enter Imaginary part of complex number:3

Enter Second complex number:
"enter real part of complex number: 4
enter Imaginary part of complex number:5

Test Cases:

2+5i, a+4i
3+6i. 6+4i
8+4i.12+7i
15-4i. 6-2i
3-8i.18-6i

Medium

1. Write a C program that uses functions and structures to perform the following operations: i). Addition of two complex numbers ii) subtraction of two complex numbers iii) Multiplication of two complex numbers iv)Division of two complex numbers

Input:
Menu for operation complex numbers
1.Addition 2.Subtraction 3.Multiplication 4.Division
2

Enter First complex number:
"enter real part of complex number: 2
enter Imaginary part of complex number:3

Enter Second complex number:
"enter real part of complex number: 4
enter Imaginary part of complex number:5

Output:
Addition of complex number :
Real part of Subtraction : -2
Imaginary part of Subtraction: -2

Test Cases:

2+5i, a+4i
3+6i. 6+4i
8+4i.12+7i
15-4i. 6-2i
3-8i, 8-6i

2. Write a C program to display the subject and mark information using Dynamic Memory Allocation for Structure

Sample Input:

Enter the number of records: 2

Enter subject 1 and marks:
Science 82
Enter subject 2 and marks:
DSA 73

Sample Output :
Science 82
DSA 73

Test Case:

Enter the number of records :4 (Any details of subject and marks)

Enter the number of records :A
Enter the number of records :1 (CPP 74.5)
Enter the number of records :1 (CPP seventy)
Enter the number of records :1 (233 75)

3. Write a C program to display the details of student(Name , Age) by passing structures to a function

Sample Input :
Enter No.Students: 1
Enter student 1 Name, Age :AAA, 25

Sample Output:
Student 1 details:
Name: AAA
Age : 25

Test Cases:

No.Student :4 (Any details of student)
No.Student: 5
No.Student: 1(62, 28)
No.Student: A
No.Student: 1(xxx, 28.2)

Tough:

1. Write a program to enter the marks of n student in four subjects using structure. Then calculate the total and aggregate, display the grade obtained by the student. If the student scores an aggregate greater than 75%, then the grade is Distinction. If aggregate is 60>= and <75, then the grade is First Division. If aggregate is 50 >= and <60, then the grade is Second Division. If aggregate is 40>= and <50, then the grade is Third Division. Else the grade is Fail.

Sample Input & Output:

Enter the marks in python: 90

Enter the marks in c programming: 91

Enter the marks in Mathematics: 92

Enter the marks in Physics: 93

Total= 366

Aggregate = 91.5

DISTINCTION

Test cases:

1. 18, 76,93,65
2. 73,78,79,75
3. 98,106,120,-95
4. 96,73, AA,95
5. 78,59.8,**,79

2. Mr. Johnson would like to know how many As, Bs, Cs, Ds, and Fs his students received on a test. He has n students who took the test. He would like to enter the student number and the number grade for the test for each student using structure. Develop the solution to print out each student's student number, number grade and the total number of As, Bs, Cs, Ds, and Fs. His grading scale is as follows: 90–100 is an A, 78–89 is a B, 65– 77 is a C, 50–64 is a D, and below 50 is an F.

Sample Input :
Enter No.Students: 1
Enter student 1 Number , Grade : 2001, A

Sample Output:
Student 1 details:
Number : 2001
Grade : A
Total no. A: 1, B:0, c=0, D=0, F=0,

Test Cases:

No.Student :4 (Any details of student Number , Grade)
No.Student: 5
No.Student: 1(2004, 28)
No.Student: A
No.Student: 1(xxx, A)